

# Yongfan(Steff) Liu

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## EDUCATION

### University of California, Irvine - Irvine, U.S.

Sept. 2020 – Present

The Henry Samueli School of Engineering — Ph.D. in Computer Engineering

### Southeast University - Nanjing, China

Sept. 2016 – June 2020

Chien-Shiung Wu (Honor) College — Bachelor of Computer Science and Technology

### RWTH Aachen - Aachen, Germany

Jan. 2018 – Mar. 2018

Germany Smart Industrial 4.0 Winter Camps Study Project

## TECHNICAL SKILLS

**Frameworks/Tools:** PyTorch, TensorFlow, OpenCV, SNPE SDK

**Research Areas:** Computer Vision, 3D Reconstruction, LLM&VLM, Edge AI, Light-weight Model, Model Quantization

**Development Platforms:** NVIDIA Orin Jetson Nano, Qualcomm Snapdragon, Qualcomm Automotive Platform

## EXPERIENCE

### Samsung Research American

Jun. 2026 - Sept. 2026

*Internship – Mobile Platform Solution, XR Platform Team*

*Mountain View, CA*

- Spatially-aware tiny VLM for egocentric AR-glasses golf videos via privileged training
- Built a 3D perception pipeline (detection, tracking, reconstruction) to recover golf ball and club-head trajectories from egocentric AR-glasses video, and used them as privileged information for a teacher VLM.
- Designed a three-round data curation pipeline (LLM validation, manual impact check, teacher generation) with a leakage gate, so the teacher's chain-of-thought supervision relies only on the images.
- Fine-tuned a lightweight Gemma 4 VLM with QLoRA and an impact-weighted loss, raising impact-detection F1 from 0% to 70% (97.2% precision, 91.4%  $\pm 1$ -frame accuracy) with image-only on-device inference.

### ILLIXR (Illinois Extended Reality) Lab

Oct. 2024 –Present

*Collaborator – Efficient Streaming 3D Reconstruction with XR devices*

*UIUC, IL & UC Irvine, CA*

- Proposed a reconstruction-aware, backbone-agnostic memory bank for VGGT-style 3D reconstruction with scene-token similarity to keep informative frames and drop redundant ones.
- Matched full-sequence accuracy using only about 3% of frames, enabled 540-frame reconstruction on a single 24 GB GPU (vs. over 60 GB), and cut StreamVGGT latency and peak memory by 4 $\times$ .
- Submitted one paper under review at ICLR 2027.

### Intelligent System Architecture Laboratory

Mar. 2023 – Nov. 2025

*Graduate Researcher – Efficient Depth Estimation with Meta Glasses*

*UC Irvine, CA*

- Proposed hardware-friendly novel efficient model with MultiHead Cost Volume, decreased inference latency by 7% and decreased error rate by 17% comparing with previous model
- Proposed Rectification Positional Encoding (RPE) to enable fast online stereo rectification, reducing inference latency by 43% and lowering the overall pipeline error rate by 42%.
- Implemented models with cross-compiling and quantization skills on Qualcomm Snapdragon devices, Nvidia Orin Jetson, and next-generation AR glasses, *Aria* .
- Published one paper to **CVPR 2025**

### NIO – Electric vehicle manufacturer

June 2024 – Sept. 2024

*Internship – Digital Cockpit Department*

*San Jose, CA*

- **Acceleration and deployment of large language models on heterogeneous in-vehicle devices**
- Optimized language model inference with speculative sampling, achieving a 2.5 $\times$  speedup.
- Deployed large language models on NVIDIA Orin Jetson Nano and Qualcomm SA-8295P platforms.
- Collaborated with other team members on generative models, including diffusion-based image synthesis.

## PUBLICATIONS & PATENTS

[C.2] **Yongfan Liu**, Boyuan Tian, Rahul Singh, Sarita Adve, and Hyoukjun Kwon: Fast and stable long-context 3D reconstruction using Recon-Aware Memory Bank. Under review of ICLR 2027

[C.1] **Yongfan Liu** and Hyoukjun Kwon: Efficient Depth Estimation for Unstable Stereo Camera Systems on AR Glasses. Accepted by **CVPR 2025**

[P.1] Brain tissue segmentation method based on diagonal voxel local binary pattern texture operator, CN110728685B . Issued 2023-04-07

[J.1] **Yongfan Liu**, Sen Du and Youyong Kong: Supervoxel Clustering with a Novel 3D Descriptor for Brain Tissue Segmentation. ACMMLC 2020 & IJMLC [ISSN: 2010-3700 (Online)]